

EC441 Midterm Two Spring 2023

This is a closed book, no calculator exam. You are permitted one legible 8.5x11 per person. This reference page must be submitted with the exam. You may complete this exam with one partner, who must be a different person than your MT 1 partner.

Q1. Subnets (25 points)

- How many '/25' networks can be subnetted from a '/28' network?
- How many '/25' networks can be subnetted from a '/20' network?
- How many hosts can be configured in a '/64' IPv6 network?
- The following code prints out nothing. What are the values that the user typed for X and for Y?

```
import ipaddress as ip

X = input()
Y = input()

a = ip.ip_network("128.197.0.0/16")

b = [ip.IPv4Network('128.197.0.0/19'), ip.IPv4Network('128.197.32.0/19'),
      ip.IPv4Network(X+"/17"), ip.IPv4Network(Y+"/18")]

for net in b:
    if not net.subnet_of(a):
        print(net, "is not a subnet of", a)

for x in b:
    for y in b:
        if x != y and x.overlaps(y):
            print(x, y, "overlap")
```

Q3. TCP (25 points)

Consider a 40 Mb/s service provided by an ISP to customer Q. This Boston-based customer connects to a website using HTTP/TCP.

The packet size is 1500 bytes (ignore the 40 bytes of TCP/IP overhead). The round trip time for this server is 0.03 seconds.

The congestion window protocol being used is slow start (starts at one packet, increases by one every ACK)

- How long will it take for the throughput of this connection not to be limited by the congestion control window?
- Does there need to be an adjustment to the receive window of Q in order to achieve a throughput of 40 Mb/s? If so, what adjustment?
- The website says it is based in San Jose, which is 4000 km from Boston. Do you believe it?

Q2. Error Control Coding (25 points)

- a. Suppose the probability of bit error on a link is 10%. Design a link-layer protocol using block codes and an error control algorithm (i.e. some combination of error correction, detection, and retransmission) that will reduce the probability of error that the network layer sees to 1% or lower.

Your code should be as short as possible: i.e. for the (n, k) block code used, you should choose to minimize the value of n .

- b. Consider the following block code, as described by these codewords and bit map:

Data Bits	Transmitted Codeword
11	00000
10	01011
01	10101
00	11110

What is the rate of the code? What is the minimum distance of the code? If the receiver receives the code word 11111, what are its possible actions?

Your friend claims to have written an algorithm using this code that simultaneously corrects all single-bit errors and also asks for retransmissions whenever there are two bit errors. Do you believe them? Why or why not?

Q4. Routing and Addressing (25 points)

- a. Design a JSON / python / pseudocode format for (1) a link state (LS) packet and (2) a distance vector (DV) packet.

Provide a network graph (make up you own connections and costs) for a network with six routers labelled A through F. Your network should have at least 8 full-duplex links.

Show an example of a LS packet and DV packet that node A would construct and send out using your packet formats defined above.

- b. Explain the following networking terms by analogy to the world-wide system of postal mail.

Here is an example answer comparing “packet” to “letter”

———— 0 (example)

packet :: letter

A postal letter is equivalent to a network packet in that

- it has a source and destination address included in a prescribed format

- it is forwarded (inspected for destination)

over several "links" through its path.

- it sometimes gets delayed or discarded inside the network

———— 1

autonomous system ::

———— 2

BGP is how inter-AS routing works ::

———— 3

OSPF is an example of how intra-AS routing works ::

———— 4

hierarchical intra-AS routing ::

———— 5

network address vs host address ::